



ARpedia allows you to pick up things in Virtual Reality and put them at different places to get more information

also performing facial recognition, i.e. associating the face to a person's identity.

- **Close-Range Depth Tracking:** The close-range depth tracking module helps with identification of things such as gesture recognition, hand/finger tracking and similar interactivity using the integrated 3D camera.
- **Augmented Reality:** This feature allows you to combine data gathered from the 3D integrated camera with a superimposed image to provide you with a virtual experience.

CREATIVE INTERACTIVE GESTURE CAMERA

The Creative Interactive Gesture Camera is a specially designed piece of hardware with depth sensitivity and microphone features. It is best used for close-range interaction. This camera is made to be coupled with the Intel Perceptual Computing SDK and can be ordered from the Intel website at http://click.intel.com/intelsdk/Creative_Interactive_Gesture_Camera_Developer_Kit-P2061.aspx.

The specifications available for this camera are as follows:

- **Maximum RGB camera resolution:** 720p
- **Maximum IR depth resolution:** QVGA (320 x 240)
- **Synchronization:** RGB + depth frame sync
- **Frame rate:** 30 fps
- **Field of View (FOV):** 73 degree diagonal
- **Range:** 0.5ft - 3.25ft
- **Microphone:** Dual array
- **Size:** 4.27in x 2.03in x 2.11in
- **Weight:** 9.56 oz
- **Power:** Single USB 2.0 (power < 2.5w)

INTERFACE GUIDELINES DESIGN PHILOSOPHY

- The software should be reality inspired but not an exact clone of reality because the virtual environment can afford to relax the rules to sometimes make things easier.
- Literal design metaphors can be used in a much better way rather than abstract symbols in a perceptual environment.
- The intuitive nature of an application is very important in helping a new user adopt it and easily get around the interface.
- Contextually appropriate interactions help ensure that the interactions match the context.

- Identifying the strengths of the new modalities of interaction helps you design according to the appropriate methods of interaction.

MINIMIZING FATIGUE

- Make the interactions less complex and easier to perform.
- Gestures should be short lasting, so that they can be performed quickly.
- Precise input should never be a mandatory requirement, as that only makes it difficult to match the expected gesture as small variations are expected to occur.
- Repeated gestures should be avoided so that the user does not get frustrated quickly.

FACE TRACKING GUIDELINES

- Expect more expressions in the SDK in the future. Currently, smiles and winks can be detected.
- Natural expressions will be difficult to detect in front of a computer. Users should be prompted to show exaggerated expressions.
- Give feedback to the user to ensure they're in a typical working distance away from the computer, for optimal feature detection.
- Give feedback to the user about any orientation or lighting issues – provide error messages or recommendations.
- For optimal tracking, have ambient light or light facing the user's face (avoid shadows).
- Try to make the interface background as close to white as possible (the screen can serve as a second light to ensure good reading).



Head of the Order allows fully integrated experience with intricate gesture recognition to conjure spells